

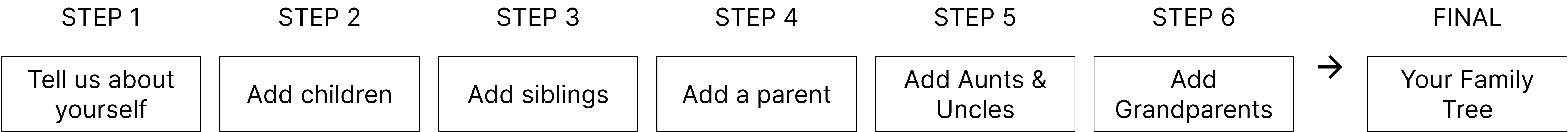


# XLH Family Tree Tool

UX & DEV LOGIC

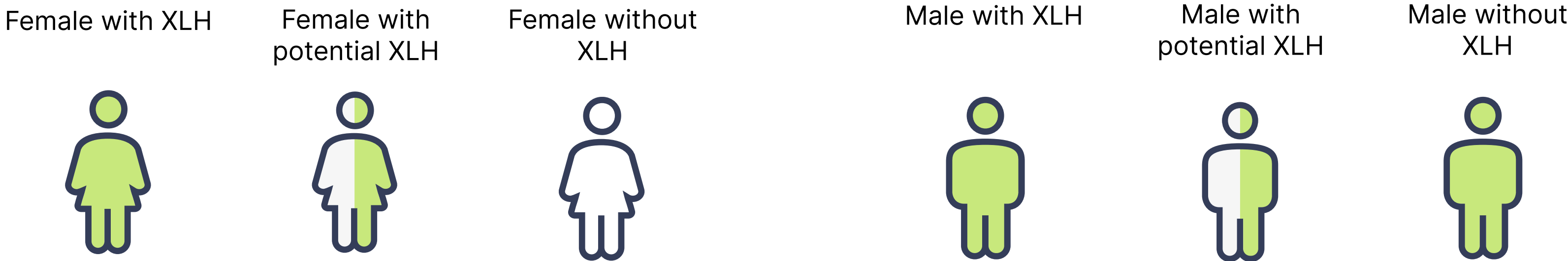
# INTRODUCTION

The XLH Family Tree Tool includes seven steps, six of which require user input, with the final step presenting the completed interactive family tree results.



# PROFILES & XLH STATUS

As users build their family tree, details like gender and XLH status are reflected in the corresponding icon.





GLOBAL LOGIC

Questions

Each step includes at least three questions.  
Any question without an asterisk is optional.

**A Do you have XLH?**  
If a user selects “Yes” or “Unsure” in response to “Do you/they have XLH?”, an additional “Add Symptoms” field will appear, allowing them to choose from a checklist of symptoms.

**B Secondary actions**  
Most steps include an optional secondary action accessible through a button within the interactive container, which opens as an overlay.

**C 50 person limit**  
Users will be limited to feature a maximum of 50 people. If a user reaches that limit, a warning will appear notifying they can either finish, or remove current members.

STEP 1 - TELL US ABOUT YOURSELF

Provide your details

\*Required field

First name

Enter name

Gender assigned at birth\*

Do you have XLH?\*

Male

Female

Yes

No

Unsure

+ Add spouse/partner

NEXT

STEP 1 - TELL US ABOUT YOURSELF

Provide your details

\*Required field

First name

Amy

Gender assigned at birth\*

Do you have XLH?\*

Male

Female

Yes

No

Unsure

Add symptoms ?

3 symptoms selected

Changes in growth and development

Bone weakness and broken bones

Joint and tendon issues

Leg and walking issues

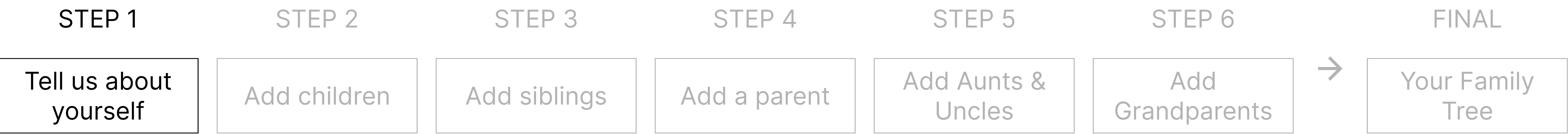
+ Add spouse/partner

NEXT

You have reached the maximum limit

The XLH family tree is limited to 50 family members.  
Please remove profiles in order to move forward.

CLOSE



Default questions

- First name
- Gender assigned at birth
- Do you have XLH?

Secondary action

- Add a partner

To consider

If a user selects “No” for themselves and “Yes” or “Unsure” for their partner’s XLH status, the family tree will shift to build out the partner’s side instead.

Questions such as ‘Add siblings’ would change to ‘Add [Partners name] siblings’

STEP 1 - TELL US ABOUT YOURSELF

Provide your details

\*Required field

First name

Enter name

Gender assigned at birth\*

☐ Male ☐ Female

Do you have XLH?\*

☐ Yes ☐ No ☐ Unsure

+ Add spouse/partner

NEXT

STEP 1 - TELL US ABOUT YOURSELF

Provide your details

\*Required field

First name

Amy

Gender assigned at birth\*

☐ Male ☒ Female

Do you have XLH?\*

☒ Yes ☐ No ☐ Unsure

Add symptoms ?

3 symptoms selected

☒ Changes in growth and development

☐ Bone weakness and broken bones

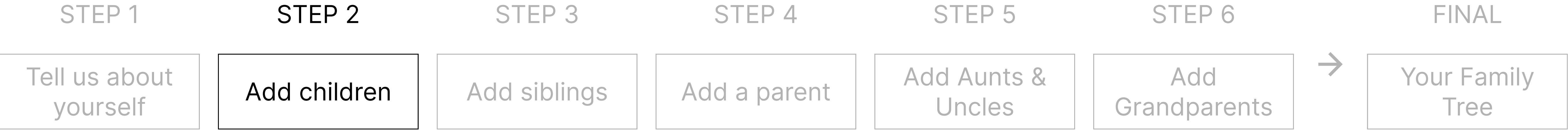
☒ Joint and tendon issues

☒ Leg and walking issues

+ Add spouse/partner

NEXT





**Optional**  
Because not all users will have children, this step is optional

- Default questions**
- First name
  - Gender assigned at birth
  - Do they have XLH?

- Secondary action**
- Add another child

STEP 2 - ADD CHILDREN

Would you like to add your children?

YES

SKIP

GO BACK

STEP 2 - ADD CHILDREN

Provide details

\*Required field

First name

David

Gender assigned at birth\*

☒ Male ☐ Female

Do they have XLH?\*

☐ Yes ☐ No ☒ Unsure

Add symptoms ?

3 symptoms selected

First name

Emily

Gender assigned at birth\*

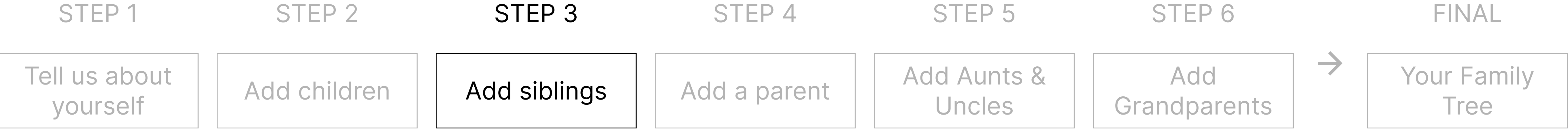
☐ Male ☒ Female

Do they have XLH?\*

☐ Yes ☒ No ☐ Unsure

GO BACK

NEXT



**Landing screen**  
Because not all users will have siblings, this step is optional

- Default questions**
- First name
  - Gender assigned at birth
  - Do they have XLH?

- Secondary action**
- Add another sibling
  - Add a siblings children (Overlay)

STEP 3 - ADD SIBLINGS

Would you like to add your siblings?

YES

SKIP

GO BACK

STEP 3 - ADD SIBLINGS

Provide your details

\*Required field

First name

Enter name

Gender assigned at birth\*

Do you have XLH?\*

☐ Male

☐ Female

☐ Yes

☐ No

☐ Unsure

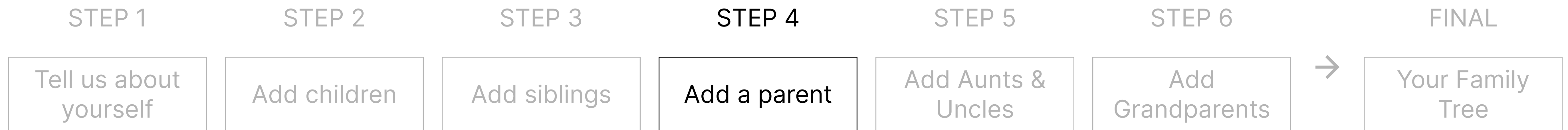
+ Add their children

+ Add sibling

GO BACK

NEXT





## Landing screen

Users will be presented with four options:

- Mom
- Dad
- I don't know
- Neither

'I don't know' & 'Neither' will transport the user to their family tree and end the experience

## Default questions

- Does your [Mom or Dad] have XLH?
- Add symptoms

## To consider

Based on the parent selected, the family tree will shift to focus on that parent's side moving forward.

STEP 4 - SELECT A PARENT

Which of your parents has or might have XLH?

MOM

DAD

NEITHER

I DON'T KNOW

?

Use the symptoms list to understand which parent may be affected by XLH. If you're still undecided, select the 'I don't know' button to end the experience and see your results.

GO BACK

STEP 4 - SELECT A PARENT

Provide details

\*Required field

Does your mom have XLH?\*

☒ Yes

☐ Unsure

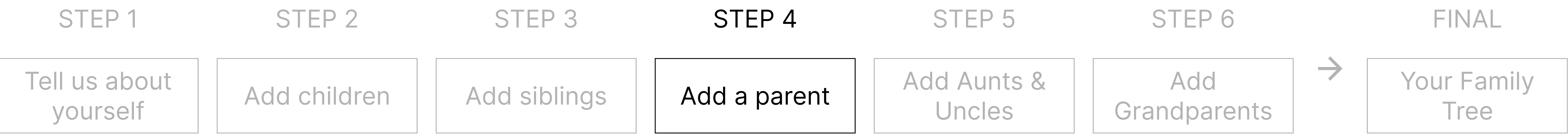
Add symptoms ?

3 symptoms selected

▼

GO BACK

NEXT



To consider

Males can only inherit XLH from their mothers. If a male user has XLH or is unsure, the ‘Dad’ button will default to a disabled state. The user will either select ‘Mom’ to move forward, or end the experience by selecting the ‘I don’t know’ / ‘Neither’ buttons.

STEP 4 - SELECT A PARENT

Which of your parents has or might have XLH?

MOM

This option isn't available. Males can only get XLH from their mom.

DAD

I DON'T KNOW

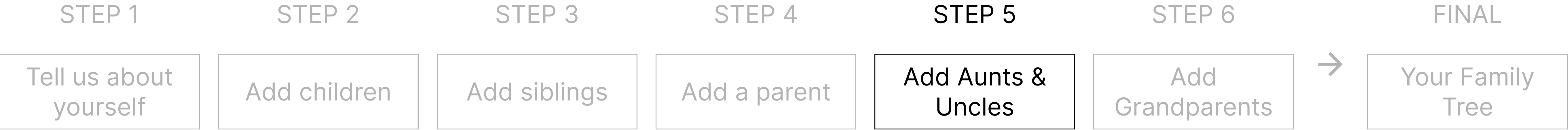
NEITHER

?

Use the symptoms list to understand which parent may be affected by XLH. If you're still undecided, select the "I don't know" button button to end the experience and see your results.

GO BACK





**Landing screen**  
Since not all users will have aunts or uncles on the selected parent’s side, this step is optional.

- Default questions**
- First name
  - Gender assigned at birth
  - Do they have XLH?

- Secondary action**
- Add another aunt or uncle
  - Add their children (Overlay)

STEP 5 - ADD AUNTS & UNCLES

Does your [mom] have brothers or sisters?

YES

SKIP

Please include biological relatives only.

GO BACK

STEP 5 - ADD AUNTS & UNCLES

Provide details

\*Required field

First name

Maria

Gender assigned at birth\*

Male

☒ Female

Do they have XLH?\*

☐ Yes

☐ No

☒ Unsure

Add symptoms ?

4 symptoms selected

+ Add their children

First name

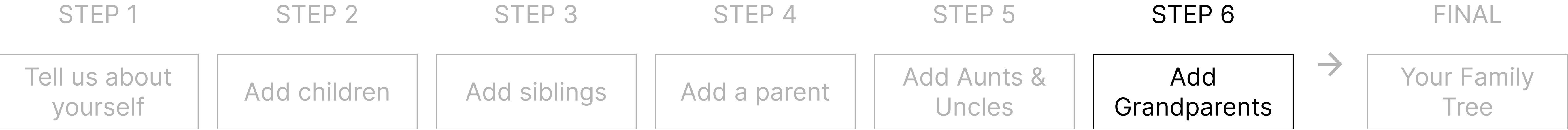
Casey

Gender assigned at birth\*

Do they have XLH?\*

GO BACK

NEXT



Landing screen

Users will be presented with four options:

- Grandmother
- Grandfather
- Neither
- I don't know

'I don't know' & 'Neither' will transport the user to their family tree and end the experience

Default questions

- Do they [Grandmother or Grandfather] have XLH?
- Symptoms

STEP 6 - ADD A GRANDPARENT

Which of your [maternal] grandparents has or might have XLH?

GRANDMOTHER

GRANDFATHER

NEITHER

I DON'T KNOW

?

Use the symptoms list to understand which parent may be affected by XLH. If you're still undecided, select the 'I don't know' button to end the experience and see your results

GO BACK

STEP 6 - ADD A GRANDPARENT

Provide details

\*Required field

Do they have XLH?\*

Yes

Unsure

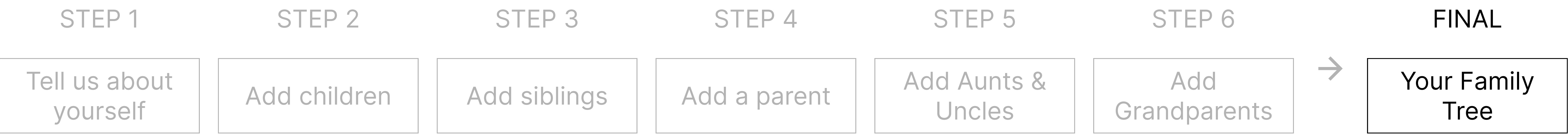
Add symptoms ?

2 symptoms selected

GO BACK

NEXT





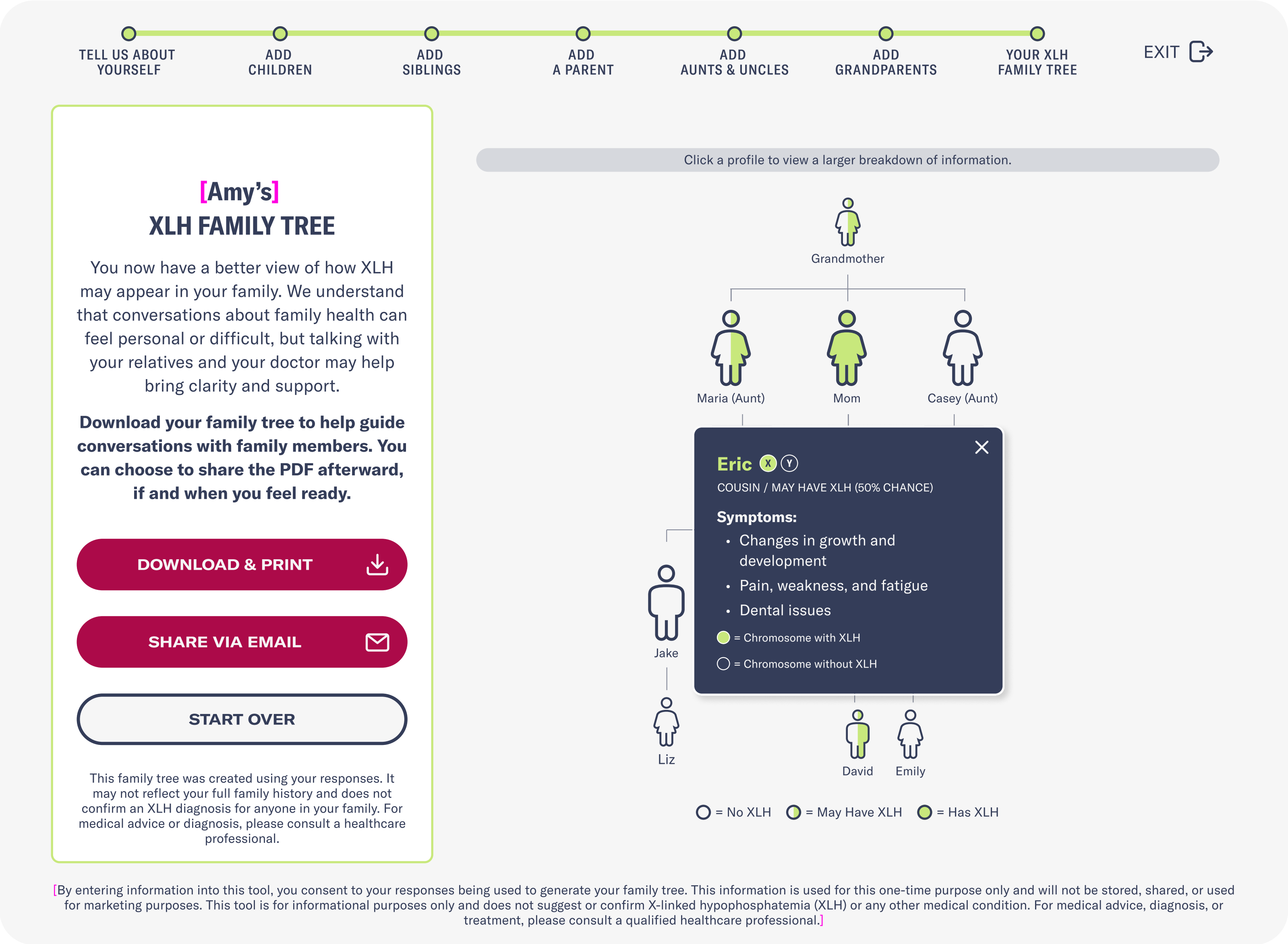
**Primary actions**

After the tree is generated, users will be able to:

- Download a PDF
- Email a PDF
- Start over

**Family tree**

Based on the information provided, users will be able to click a profile and view information via an overlay



STEP 1

STEP 2

STEP 3

STEP 4

STEP 5

STEP 6

FINAL

Tell us about  
yourself

Add children

Add siblings

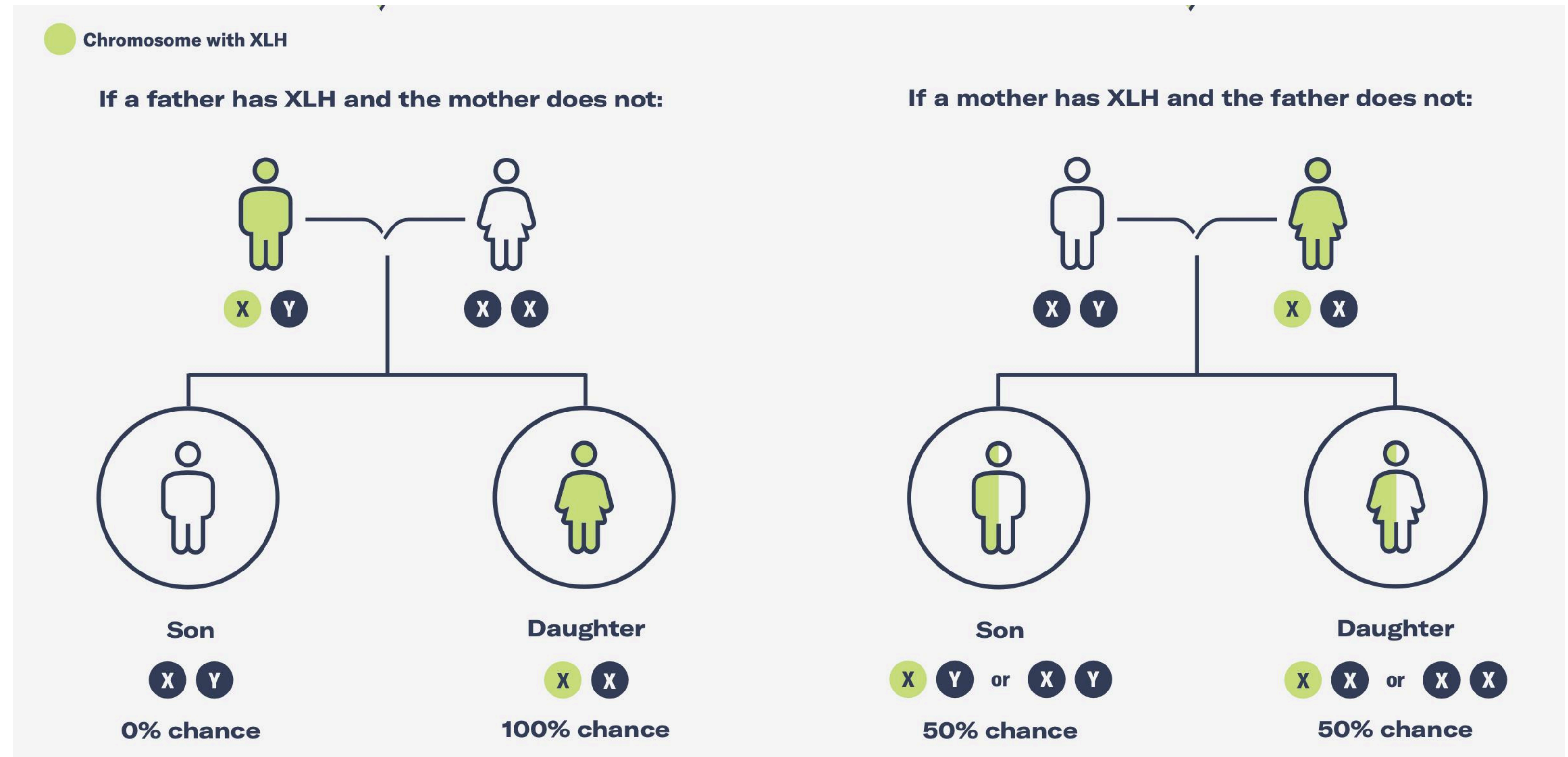
Add a parent

Add Aunts &  
UnclesAdd  
GrandparentsYour Family  
Tree

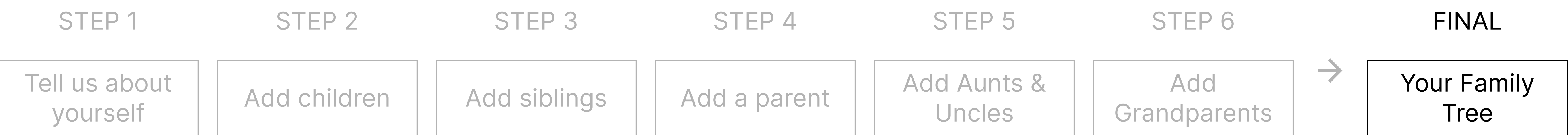
## Chromosomes

XX=Female / XY=Male

- If users have or may have XLH, one chromosome will be highlighted in green.
- Since males can only inherit via their mother, their 'X' chromosome will be highlighted.







PDF

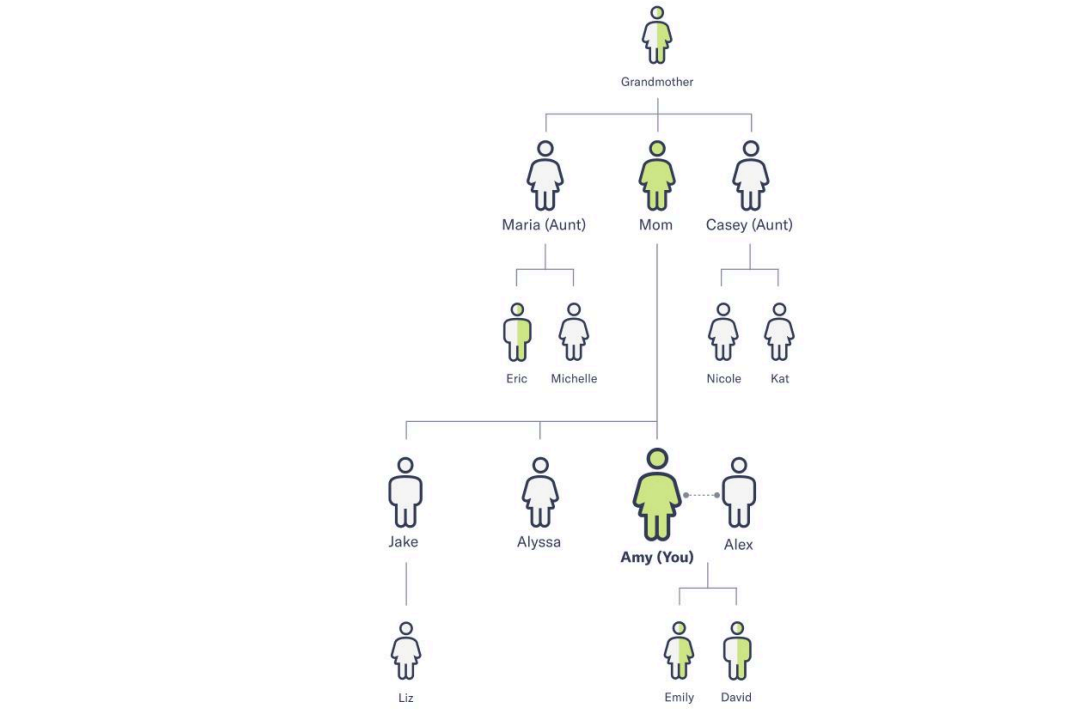
The generated PDF is broken down into 3 pages or sections

- Visual tree
- Table chart of information
- List of symptoms

AMY'S XLH FAMILY TREE

You now have a better view of how XLH may appear in your family. We understand that conversations about family health can feel personal or difficult, but talking with your relatives and your doctor may help bring clarity and support. You can share this PDF to help guide conversations with family members, if and when you feel ready.

This family tree was created using your responses. It may not reflect your full family history and does not confirm an XLH diagnosis for anyone in your family. For medical advice or diagnosis, please consult a healthcare professional.



XLH

LINK

Legend: No XLH, May Have XLH, Has XLH

PAGE 1 OF 4

AMY'S XLH FAMILY TREE

This family tree was created using your responses. It may not reflect your full family history and does not confirm an XLH diagnosis for anyone in your family. For medical advice or diagnosis, please consult a healthcare professional.

| NAME     | FAMILY MEMBER | XLH STATUS*                | SYMPTOMS                                                                                                        |
|----------|---------------|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| AMY      | Self          | Has XLH* (100%)            | • Changes in growth and development<br>• Joint and tendon issues<br>• Leg and walking issues                    |
| ALEX     | Partner       | No XLH                     | N/A                                                                                                             |
| DAVID    | Child         | May have XLH* (50% chance) | • Bone weakness and broken bones<br>• Joint and tendon issues<br>• Hearing issues                               |
| EMILY    | Child         | No XLH* (5% chance)        | N/A                                                                                                             |
| JAKE     | Sibling       | No XLH                     | N/A                                                                                                             |
| LIZ      | Niece         | No XLH                     | N/A                                                                                                             |
| ALYSSA   | Sibling       | No XLH                     | N/A                                                                                                             |
| MOM      | Mother        | Has XLH* (100%)            | • Bone weakness and broken bones<br>• Dental issues<br>• Hearing issues                                         |
| MARIA    | Aunt          | May have XLH* (50% chance) | • Changes in growth and development<br>• Leg and walking issues<br>• Joint and tendon issues<br>• Dental issues |
| ERIC     | Cousin        | May have XLH*              | • Bone weakness and broken bones<br>• Pain, weakness, and fatigue<br>• Dental issues                            |
| MICHELLE | Cousin        | No XLH                     | N/A                                                                                                             |

XLH

LINK

Legend: Chromosome with XLH, Chromosome without XLH

\*Calculated based on user response. Not a formal diagnosis.

PAGE 2 OF 4

Not everyone experiences XLH the same way.

You and your family member could have several of the same symptoms, or only a few. Reviewing the symptoms below may be helpful when considering if XLH has affected your family member.

**SYMPTOMS**

**Changes in growth and development**

- Shorter than average height
- Weakening of growing bones (rickets)

**Joint and tendon issues**

- Joint damage (osteoarthritis)
- Hardening of tissue attaching bones and muscles (enthesopathy)

**Pain, weakness, and fatigue**

- Bone and joint pain
- Muscle pain and weakness
- Fatigue

**Dental issues**

- Issues such as abscesses and tooth loss
- Problems with the gums, like redness, swelling, or infection (periodontitis)

**Bone weakness and broken bones**

- Weakening of mature bone (osteomalacia)
- Broken bones (fractures) and areas of weakened bone that are not completely broken (pseudofractures)

**Leg and walking issues**

- Delayed walking or an unusual way of walking (gait abnormalities)
- Bowed legs and knock knees

**Head and spine complications**

- Unusual shape of the head (craniosynostosis)
- Narrowing of spaces in the spine (spinal stenosis)

**Hearing issues**

- Hearing loss



Could XLH be part of your story, too?

Scan to learn more about getting an XLH diagnosis and how to start a conversation with your doctor.

XLH

LINK

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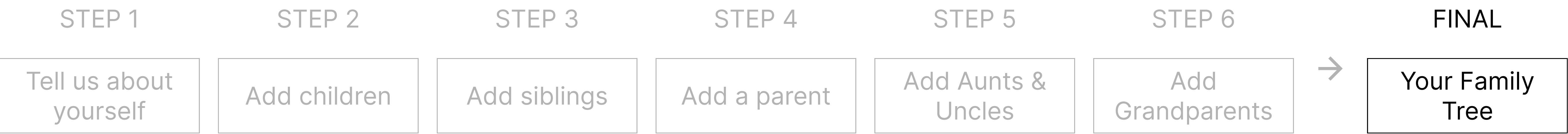


Table chart

The table chart is segmented into 4 categories:

- Name
- Family member
- XLH status
- List of symptoms

XLH Status

Status percentage is only calculated for direct offspring from users who answered ‘Yes’ to ‘Do you have XLH?’

Please view the logic chart on the next page

AMY'S XLH FAMILY TREE

This family tree was created using your responses. It may not reflect your full family history and does not confirm an XLH diagnosis for anyone in your family. For medical advice or diagnosis, please consult a healthcare professional.

XLH

FAMILY TREE

| NAME     | FAMILY MEMBER | XLH STATUS*                                                                                       | SYMPTOMS                                                                                                                                                   |
|----------|---------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AMY      | Self          | <div><div></div><div></div></div> Has XLH* (100%)                                                 | <div><div>• Changes in growth and development</div><div>• Joint and tendon issues</div><div>• Leg and walking issues</div></div>                           |
| ALEX     | Partner       | <div><div></div><div></div></div> No XLH                                                          | N/A                                                                                                                                                        |
| DAVID    | Child         | <div><div></div><div></div></div> OR <div><div></div><div></div></div> May have XLH* (50% chance) | <div><div>• Bone weakness and broken bones</div><div>• Joint and tendon issues</div><div>• Hearing issues</div></div>                                      |
| EMILY    | Child         | <div><div></div><div></div></div> No XLH* (0% chance)                                             | N/A                                                                                                                                                        |
| JAKE     | Sibling       | <div><div></div><div></div></div> No XLH                                                          | N/A                                                                                                                                                        |
| LIZ      | Niece         | <div><div></div><div></div></div> No XLH                                                          | N/A                                                                                                                                                        |
| ALYSSA   | Sibling       | <div><div></div><div></div></div> No XLH                                                          | N/A                                                                                                                                                        |
| MOM      | Mother        | <div><div></div><div></div></div> Has XLH* (100%)                                                 | <div><div>• Bone weakness and broken bones</div><div>• Dental issues</div><div>• Hearing issues</div></div>                                                |
| MARIA    | Aunt          | <div><div></div><div></div></div> OR <div><div></div><div></div></div> May have XLH* (50% chance) | <div><div>• Changes in growth and development</div><div>• Leg and walking issues</div><div>• Joint and tendon issues</div><div>• Dental issues</div></div> |
| ERIC     | Cousin        | <div><div></div><div></div></div> OR <div><div></div><div></div></div> May have XLH*              | <div><div>• Bone weakness and broken bones</div><div>• Pain, weakness, and fatigue</div><div>• Dental issues</div></div>                                   |
| MICHELLE | Cousin        | <div><div></div><div></div></div> No XLH                                                          | N/A                                                                                                                                                        |

XLH

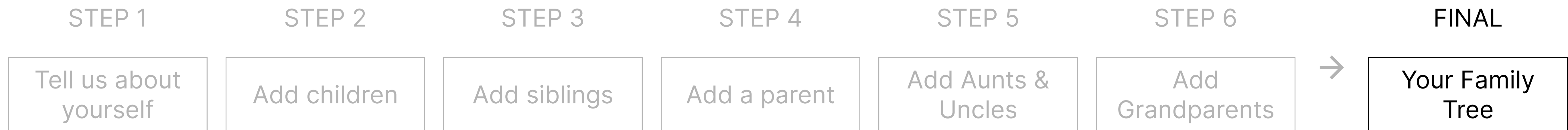
LINK

= Chromosome with XLH  = Chromosome without XLH

\*Calculated based on user response. Not a formal diagnosis.

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### XLH status percentage calculation

Answered 'yes' to 'Do you have XLH?'

